Factors influencing housing price using multiple linear regression

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***Abstract*—The price of the house is dependent on number of parameters and due to which the price fluctuate fluctuates. The aim of the study is to build a model that predicts the house price and becomes a business decision for customers. This study involves building a regression model for house price prediction and concludes by validating the Gauss Markov assumptions of linear regression, if the model built on the sample data is an unbiased estimator of the dependent variable.**

I. INTRODUCTION

Prediction helps in better decision making of a situation and estimating it effectively saves time, money and leads to take an informed decision. There have been studies carried out to predict certain event likely to happen based on certain assumptions. One of the tasks in machine learning is prediction. Predictive approach carried by statisticians or scientists is the regression analysis technique to predict based on the data collected for analysis.

Housing price, in a similar way can be predicted using parameters that highly contribute to the costlier pricing of apartments. In the dataset, the dependent variable Sale Price may depend on certain independent variables and using regression, models can be built with required features that effectively predict the pricing. The study deals with creating a model for housing dataset with multiple linear regression and test the assumption of Gauss Markov.

II. DATA EXPLORATION

The dataset contains eighteen features including the dependent variable Sale price. The independent variables available to predict the price contain both discrete and continuous data.

A. Numerical Data

The independent variables having numerical data with measurement of feet/ square feet is lot frontage, lot area of the house, total size of basement in square feet, size of first floor in square feet, size of second floor in square feet. Other set of variables available contains the count of specific amenity in the house that drives the price like number of full baths, half baths, number of bedrooms above ground, kitchens above ground and count of fireplaces available in the house.

Latitude and Longitude coordinate of the house is present in dataset since proximity close to particular locations will drive the price both ways. Years built is an interval data which signifies the age of the building.

Sale Price is another numerical data which is the dependent variable in our study.

B. Categorical Data

Dataset provided contains both Nominal and ordinal categorical data.

Ordinal level of measurement is provided in the feature Overall condition of the house ranging from very poor to excellent and the other feature External condition varies from Poor to Excellent.

Nominal data is used in Building type and house style which categorises the style like a one-storey or two-storey house and building type such as Duplex, One family, Townhouse, etc.

Code artifact written in python is used for the preliminary check in data for identifying any missing or null values and use specific technique in case for missing data.

The dataset is split into train and test for building regression model in the ratio 80:20 with random seed of student number in this study.

III. EXPLORATIORY DATA ANALYSIS

Preliminary analysis carried out in training data to understand the distribution and spread of the data for both categorical and numerical datatypes.

Analysing dependent variable Sale price ranges more than 700,000 and is right skewed as looking at the distribution the sample contains data majority between 100,000 and 400,000.

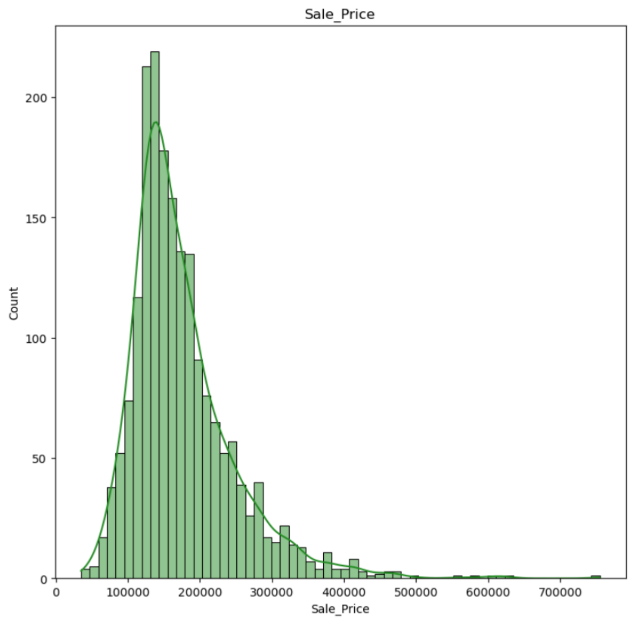

Fig. 1. Histogram on Sale Price of house in training data.

Histogram and Boxplot were used to visualize the spread of values in training data to identify skewness or presence of outliers in numeric and categorical feature of training dataset.

Analysing categorical data, majority of the sample has ‘Average’ overall condition and ‘Typical’ external condition.

The average sale price of house due to overall condition shows a significant raise in house price when moved from very poor quality to Excellent condition house.

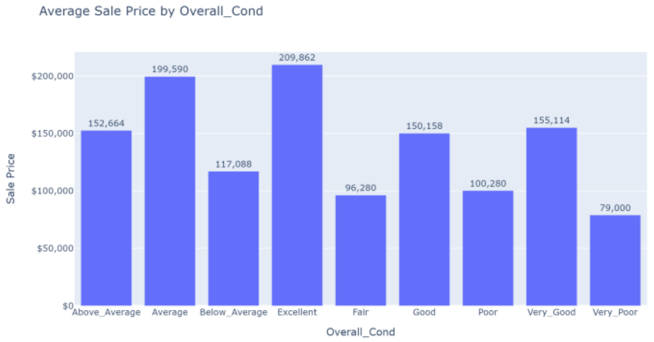

Fig. 2. Bar chart on average distribution of Sale price by oveall condition

Similarly, the average house price based on house type shows a higher price for townhouse and one family house. Two family converted is having least average price in the sample training data.

Calculating the age of the house and analysing its relation to the house price shows, price is relatively low for older houses and could notify certain newer house recently built are costlier in the dataset. It can be noted that majority of the observations in the sample have house price lower than 500,000 and few distinct houses are costly representing a smaller proportion in the dataset.

Fig. 3. Scatterplot on Sale Price vs Age of the house

Latitude and longitude coordinates of the house in sample is distributed in a smaller compact area and so transformation of coordinates is not applied. Its influence on the sale price is studied and the costlier house are located very near to each other.

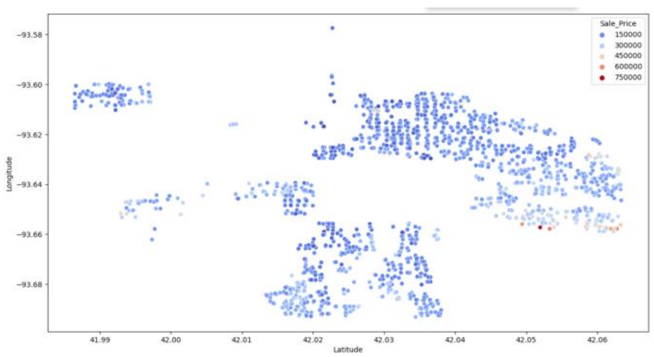

Fig. 4. Scatterplot on geological coordinates and house price

To understand the linear association between numerical variables, Pearson correlation coefficients are calculated and could identify certain independent variables showing strong positive correlation with the sale price. One feature noted is the 'Area of First floor' which shows a linear association with the response variable. In real world, it is quite significant since the price of the house depends on the size of the living area. There is a linear association present in the dataset.

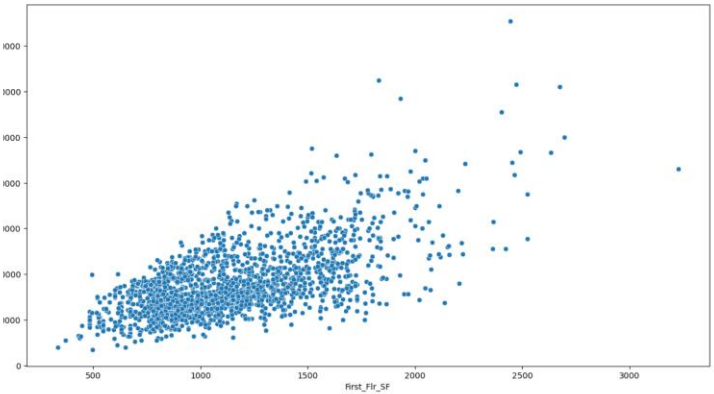

Fig. 5. Scatterplot on First_Flr_SF vs house price

Analysing the numerical variables, distribution of the data is plotted in histogram and majority of the features are right skewed and necessary transformations will be required to apply during the modelling phase.

IV. MODELLING AND DIAGNOSTICS

The intent of the paper is to investigate the factors influencing the pricing of house and for which Multiple linear regression approach is chosen. The model built using regression helps to predict housing prices and the paper comments on the Assumptions for multiple regression whether the model designed satisfies all the assumption of Gauss Markov.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + .. + \beta_n X_n + \varepsilon$$

To achieve this, sample housing data is taken and from which the sample statistic calculated will be an unbiased estimator of the population parameters. This is the foundation that regression model is built upon assuming data collected is from a normal distribution, presence of linear association between the dependent and independent variables, no multicollinearity, Zero mean and constant variance- homoscedasticity, errors are distributed normally. If all these conditions were met, the regression method modelled is the Best linear unbiased estimator for the population.

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + .. + \beta_n X_n$$

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2 + .. + \beta_n X_n$$

$$\hat{Y} \ - Predicted\ value\ of\ the\ dependent\ variable$$

$$b_0, b_1, b_2, .. b_n\ are\ the\ estimates\ of\ \ \beta_0, \beta_1, \beta_2, \dots \beta_n$$

In this study Multiple linear regression using Ordinary least square method (OLS) is used in python to estimate the housing prices.

Empty, Null, and missing values were handled during preprocessing and checks were done to find missed data. The provided sample data don't have any missing or null values.

Sample datasets may carry outliers as it might be captured wrongly, and it may be from a different distribution. Checking and removing of outliers is essential to represent a normal distribution of sample data, so model built from the dataset will not be impacted due to outliers and any influential points.

In this study, outliers are handled by calculating Z-Score for the variables and with a threshold of 3 standard deviations, data above this barrier is removed from the training data.

$$Z = X - \mu / \sigma$$

Plotting Boxplot, and histogram helps to further visualize the presence of outliers after removal.

In this stage, Kitchen_AbvGr variable has been dropped from the model as the count of kitchen above ground-floor is same for all the observations in training dataset.

The ordinal categorical variables, Overall condition, and External condition, due to natural order inbuilt in these fields, the textual fields are changed to numerical ranking orders corresponding to the conditions of the house. For the other nominal categories, dummies have been used for building type and house style. Since there is no significant order in these features, a binary classifier – dummies is encoded to the features where 1 signifies the presence of the particular variable and 0 represents the absence of this variable. One of the columns from each variable created by dummies in python is dropped as to avoid the dummy variable trap. This is carried out to avoid the occurrence of multicollinearity as the third variable can be predicted from the values other two variables in the model.

The dependent variable Sale price has majority of the value is concentrated between 100,000 and 300,000 and due to which the training data is right skewed. Square root transformation has been applied to this variable to simulate the data is from a normal distribution.

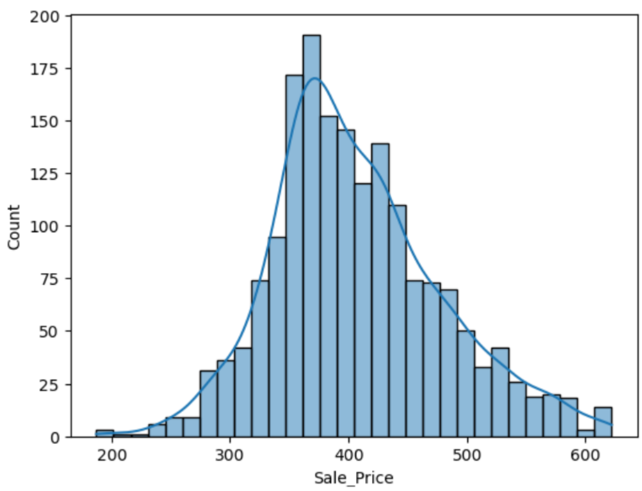

Fig. 6. Histogram of Sale Price after transformation

Statsmodel OLS regression function is used in this estimation and the dependent variable and regressors are supplied to the function and the model summary is generated from training dataset.

A. *Interpretation of initial model summary*

The Statsmodel OLS function builds a linear regression model based on data provided and significance level of 5% (α of 0.05) is used throughout the analysis. The model coefficients were calculated for each of the regressors. The initial model built has an R squared of 0.868, which tells that 86.6% of proportion of variability in the housing price is explained by the independent variables taken for this model. The remainder is the unexplainable error part.

The initial model built, has an F-statistic of 461.0 which is the measure of the ratio of Mean square of the regression to the mean square of the errors. The probability of this F-statistic is 0.00 which is less than the α of 0.05, inferring the model is statistically significant.

For testing the significance of regression coefficients calculated from the initial model, individual t-statistic along with the probability of the t statistic needs to be verified. Here null hypothesis will be the regressor coefficient is Zero and the alternate hypothesis is non-Zero.

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

$$where\ \beta_1\ is\ a\ regression\ coefficient$$

For few of the independent variables in the initial model, the probability of t-statistic is greater than 0.05 and hence they are in the non-rejection zone meaning fail to reject null hypothesis. Variables having probability of t- statistic > 0.05 were dropped as they are not significant to the model.

Building another intermediary model using fewer independent variables that has significant t value from the previous model is considered and a model with adjusted R square of 0.866 is reached. The small increase in this value is due to the reduction in the degrees of freedom in the model.

B. *Check for Correlation and Multicollinearity*

Pearson correlation coefficient is calculated in python and a plot on heatmap is generated to understand the degree of relatedness among the independent and dependent variables. On checking for multicollinearity among independent variables, Total basement squarefeet and First floor squarefeet are strongly correlated to each other. The usage of these two variables in the model will lead to multicollinearity and total basement squarefeet is dropped from the model.

To verify any further presence of multicollinearity between the variables, Variance Inflation Factor is calculated, and the values are less than 5 for all the independent variables. Presence of variables having VIF of 5-10 have considerable multicollinearity in the model and above 10 shows strong multicollinearity.

C. *Interpreatation of Final model*

The final model is built after dropping all insignificant variables from initial model.

Probability of F-statistic signifies the overall model is statistically significant and the t-statistic of the individual regressors are also significant.

For testing the overall significance of the final model, overall F test is carried out where the null hypothesis states that all the regression coefficients are equal to zero whereas the alternative hypothesis is atleast one of the regression coefficients is non-zero.

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \ldots = \beta_k = 0$$

$$H_a : Atleast\ one\ of\ the\ regression\ coefficients\ is \neq 0$$

The probability of F statistic is < 0.05 and hence the null hypothesis is rejected, and it indicates that atleast one of the independent variables is adding significant predictability for Y- Sale Price.

TABLE I. REGRESSION COEFFICIENTS FROM FINAL MODEL

Regressors	Coefficients (βs)
Const	-1.36E+04
Lot_Frontage	0.0811
Lot_Area	0.001
First_Flr_SF	0.1289
Second_Flr_SF	0.0904
Full_Bath	4.1579
Bedroom_AbvGr	-11.5756
Fireplaces	8.8016
Longitude	-83.9727
Latitude	140.7516
Age	-1.0746
Overall_Cond_Num	13.2844
Twnhs	-19.5884

Similarly, individual t test is carried out with the t-statistic for the regressors in the model and each of the independent variable in the final model are statistically significant in the model.

D. *Interpreting coefficients*

The regression coefficients of the final model is in table 1.

By interpreting, A unit increase in First floor square feet will increase the Sale Price of the house by a factor of 0.1289 provided other regressors remain constant similarly an increase in the area of second floor square feet will increase the sales price by a factor of 0.0904.

A unit increase in the count of Full bath in a house will drive the house price by a factor of 4.1579 and a unit increase in the count of Fireplaces will increase the house price by 8.8016, provided other variables remain constant.

When the age of the house increases by a year, the price of the house decreases by a factor of 1.0746 and depending on the overall condition number of the house, the price of the house increase by a factor of 13.2844, provided other variables remain constant.

Interpreting of certain coefficients in the model might not be quite reliable in real world like where the constant is -1.36E+04 meaning when regressors remain constant, the price is 1.36E+04, which is not reliable in actual world scenario.

Equation of this multiple regression model arrived is

Y= -1.36E+04 + 0.0811 (Lot_Frontage) + 0.001 (Lot_Area) + 0.1289 (First_Flr_SF) + 0.0904 (Second_Flr_SF) + 4.1579 (Full_Bath) - 11.5756 (Bedroom_AbvGr) + 8.8016 (Fireplaces) - 83.9727 (Longitude) + 140.7516 (Latitude) - 1.0746 (Age) + 13.2844 (Overall_Cond_Num) - 19.5884 (Twnhs)

V. DIAGNOSTICS

To find the fit of the regression model, residual analysis is carried out to test the underlying Gauss Markov assumptions for linear regression.

The assumptions in linear regression are.

1) *The model is linear.*
2) *The error terms have constant variances*
3) *The error terms are independent.*
4) *The error terms are normally distributed.*

A. *Linearity check*

The relationship between independent and dependent variables has a linear association between each other and it can be seen from the overall F test and individual t test, there is a linear relationship between them.

B. *Constant variance (Homoscedasticity)*

Variance Inflation Factor is calculated during the modelling process and the value is in the acceptable range of 1 to 5.

Residual vs Fitted plot is plotted to check the distribution of residuals and the residuals are distributed in random around the line. We can conclude homoscedasticity holds.

Breusch Pagan test is conducted for the presence of constant variance in the final model.

H_0: *Homoscedasticity in model showing constant error variance*

H_a: *Heteroscedasticity in model showing constant error variance*

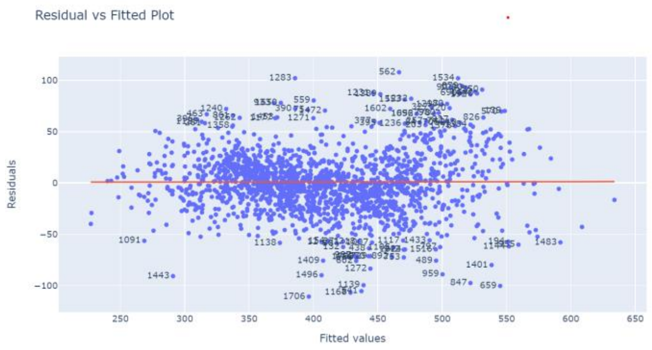

Fig. 7. Scatterplot on Residual vs Fitted

The p value calculated from the test is less than α and hence null hypothesis is rejected, and the residuals do not exhibit constant variance.

There might be presence of other correlating features in the final model that causes a non-constant variance in the model.

C. *Normality check*

Quantile-Quantile plot was drawn to find the distribution of residuals if normally distributed.

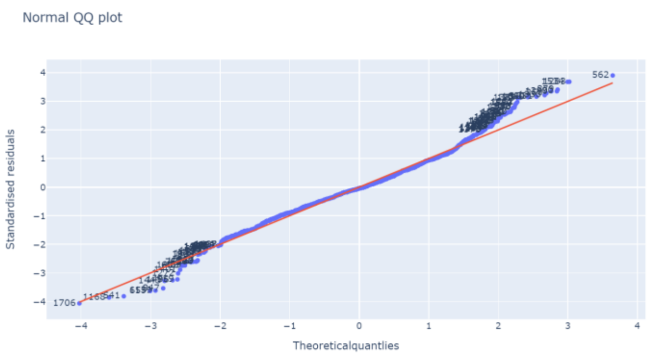

Fig. 8. Normal QQ plot

On analyzing the plot, most of the residuals are around the line showing the residuals are normally distributed. Few of the residuals are located at a distance from the line.

The Shapiro Wilk test is carried out and the probability calculated is less than α.

$$H_0 : Sample\ drawn\ from\ normal\ distribution$$

$$H_a : Sample\ drawn\ from\ non-normal\ distribution$$

Hence the null hypothesis is rejected and can conclude the sample is from a non-normal distribution.

D. Independent error terms

To verify the presence of autocorrelation between the residuals, The Durbin Watson test is calculated and a statistic of 1.959 is achieved. This shows that there is no autocorrelation among the residuals.

Only few of the Gauss Markov assumptions like independent error terms, and linearity were able to be verified using the residuals. However, due to the failure of other assumptions the sample data shows heteroscedasticity implying non constant variance which might be due to presence of outliers and the sample is from a non-normal distribution.

VI. EVALUATION

With the final model derived, using the testing data, similar transformations are applied to the dataset and insignificant columns were dropped. Fitting the training data, the predicted sale price is plotted against the actual sale price. The prices are linearly distributed around the regression line.

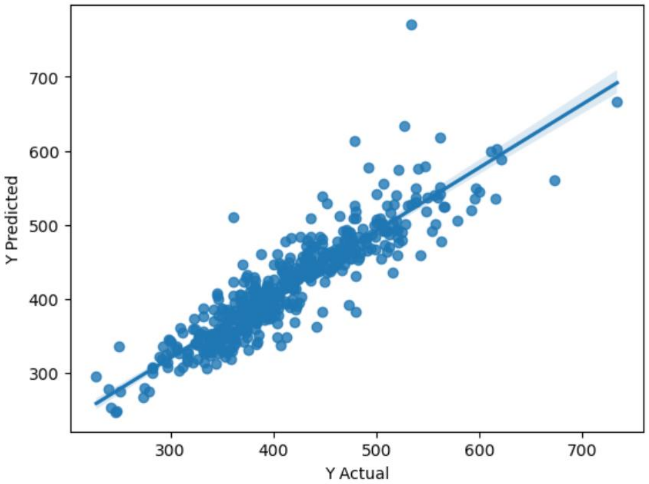

Fig. 9. Scatterplot on Y-predicted vs Y-actual

A. Examination of regression output

From the overall model, the coefficient of determination, R square arrived is 0.849 and adjusted R square is 0.848.

84.9% of variability in the sales price is explained by the independent variables in the model.

The adjusted R square explains 84.8% of variability in Sale Price is explained by the independent variables with 12 degrees of freedom. This adjusted value is more reliable compared to actual R square

Mean Absolute Error for this model is 23.0423 and Mean Absolute Percentage Error is 0.056

The average difference between the actual and the forecasted value by the model is 5.6%.

On calculating from the residuals, Mean Squared error and root mean squared error is 1081.21 and 32.882 respectively.

If further models were built and using the test dataset, the accuracy of a specific model having best can be picked from MSEs having least value.